

Project Overview: Data Cleaning & Feature Engineering

1. Project Background

In real-world business environments, data is rarely clean or analysis-ready. Organizations often deal with datasets that contain missing values, duplicate records, outliers, inconsistent formats, and incorrect data types. This project simulates such a real-world scenario using a large retail sales dataset to demonstrate practical data cleaning and feature engineering techniques.

2. Objective

The objective of this project is to transform a raw retail sales dataset containing over 300 records into a clean, structured, and analysis-ready dataset. The project focuses on improving data quality and usability rather than building models, as clean data forms the foundation of all successful analytics and machine learning solutions.

3. Dataset Description

The dataset represents retail sales transactions and includes attributes such as order details, customer demographics, product categories, quantity purchased, and pricing information. The raw dataset intentionally includes common data quality issues such as missing demographic values, duplicate transactions, extreme outliers, and inconsistent categorical labels.

4. Data Cleaning Process

The data cleaning process involved multiple steps: identifying and handling missing values using statistical imputation methods, removing duplicate records to avoid skewed analysis, correcting data types for accurate computations, standardizing categorical variables, and detecting and treating outliers using the Interquartile Range (IQR) method.

5. Feature Engineering

Feature engineering was performed to enhance the analytical value of the dataset. New features such as total sales per order and date-based attributes (year, month, weekday) were derived from existing columns. These features enable deeper business insights, trend analysis, and support downstream tasks such as dashboarding and predictive modeling.

6. Final Outcome

The final cleaned dataset is free from major inconsistencies and is well-structured for analysis. It is suitable for exploratory data analysis, business intelligence dashboards, SQL-based analytics, and machine learning workflows. This project demonstrates strong practical skills in data preprocessing and reflects real-world data handling experience.